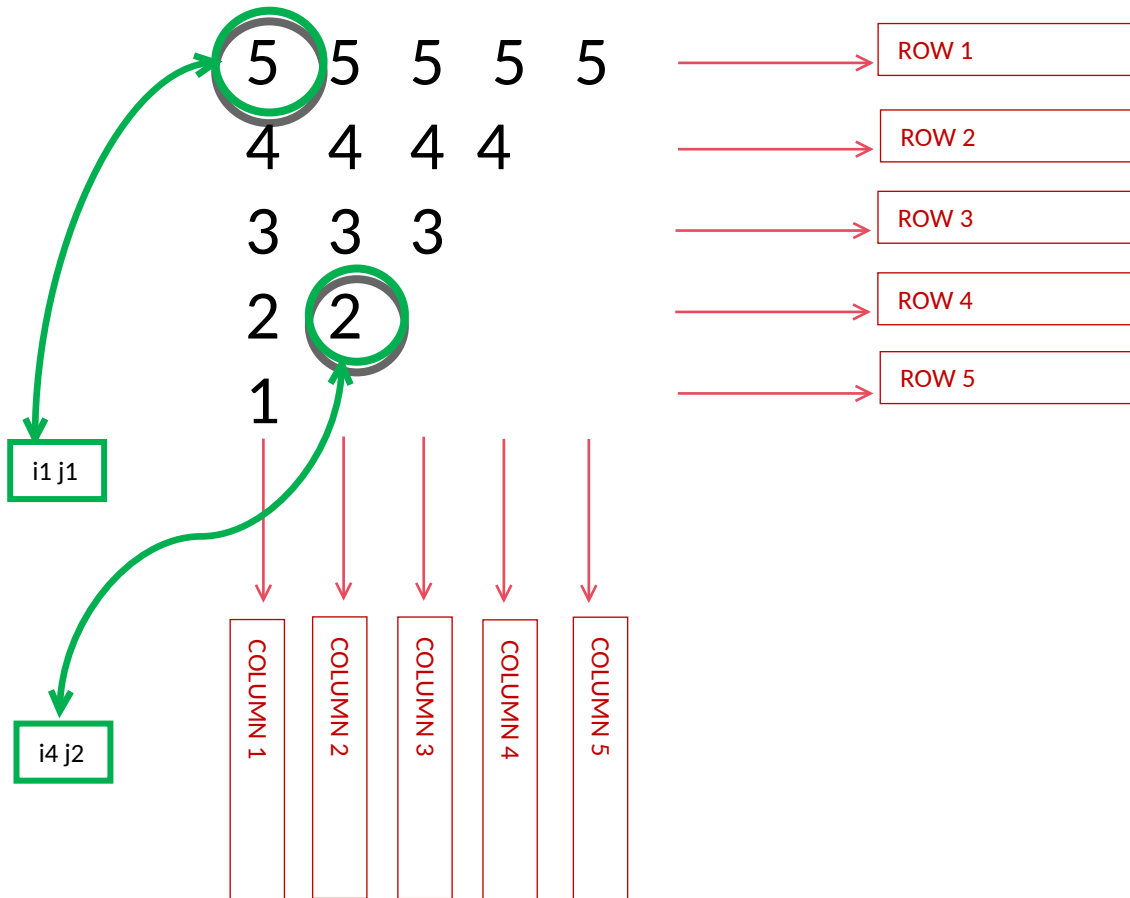


WEEK 01 MODULE
Programming Practice

TOPIC : 2 'Number Pattern'

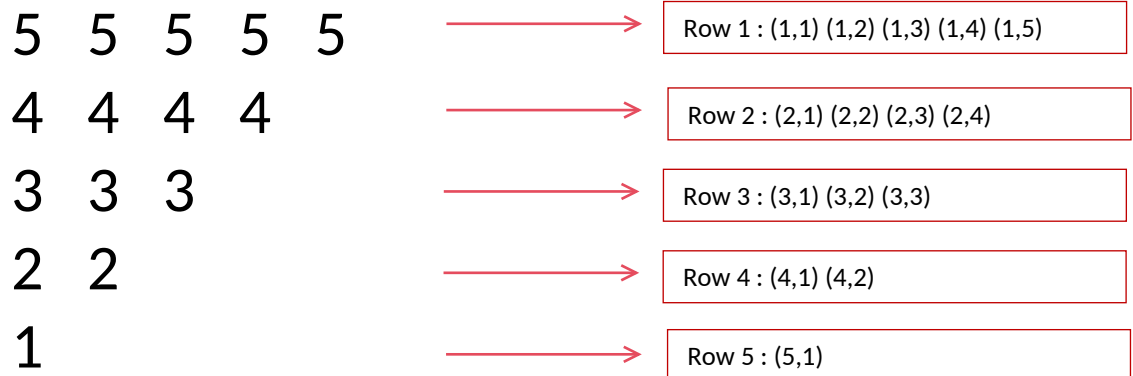
5. INVERTED SAME NUMBER TRIANGLE



1. Problem Analysis :-

i. Pattern Structure:

- The output forms an inverted same numbered right-angled triangle.
- This pattern is also based on the **Right Angled Triangle structure**, but inverted.
- Unlike the Number Triangle, here the number of elements **decreases row by row**.
- each row prints the **same number repeatedly**, and that number is equal to the row index.
- For n=5



ii. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)
- size = $n \times n$.
- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.
- For $n = 5$:

Observation:

Row 1 → 5 5 5 5 5
 Row 2 → 4 4 4 4
 Row 3 → 3 3 3
 Row 4 → 2 2
 Row 5 → 1

A clear relationship can be observed:

- Number of elements decreases each row
- All elements in a row are identical
- Value depends on reverse row index

- Value formula: $n - i + 1$

Step 2: Traverse the grid row by row using outer loop.

- The outer loop is responsible for generating rows.
- for (i = 1 to n)

Step 3: For every row, to control shrinking structure.

- for (j = 1 to (n-i+1))

Step 4: Print value = n-i+1

Step 5: After all numbers of the current row have been printed, move the cursor to the next line.

Step 6: Repeat the same process until all n rows have been completed and pattern shrinks to 1 element

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class InvertedSameNumberTraingle {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++) {

            for(int j = 1; j <= (n - i + 1); j++) {
                System.out.print((n - i + 1) + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

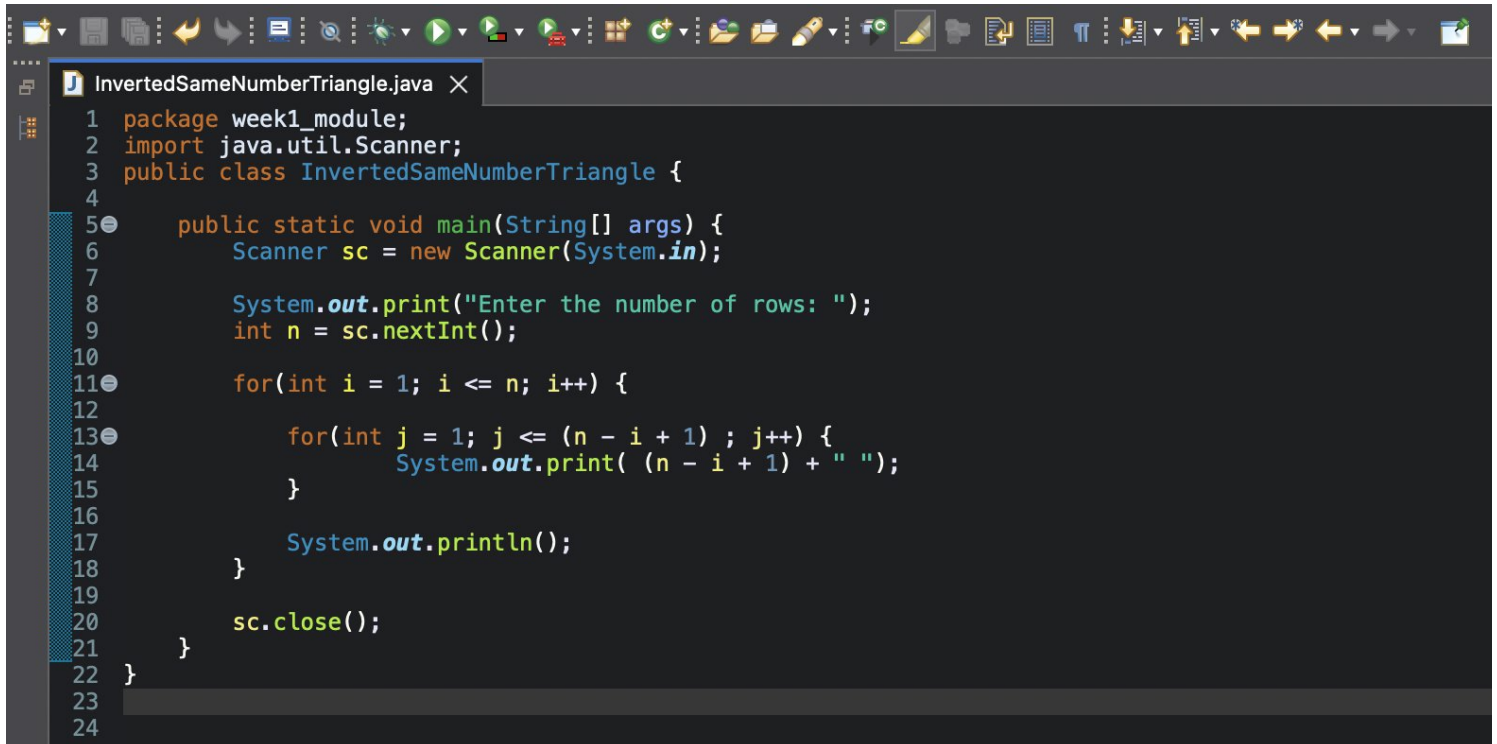
3. Program Output :-

Enter the number of rows: 5

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

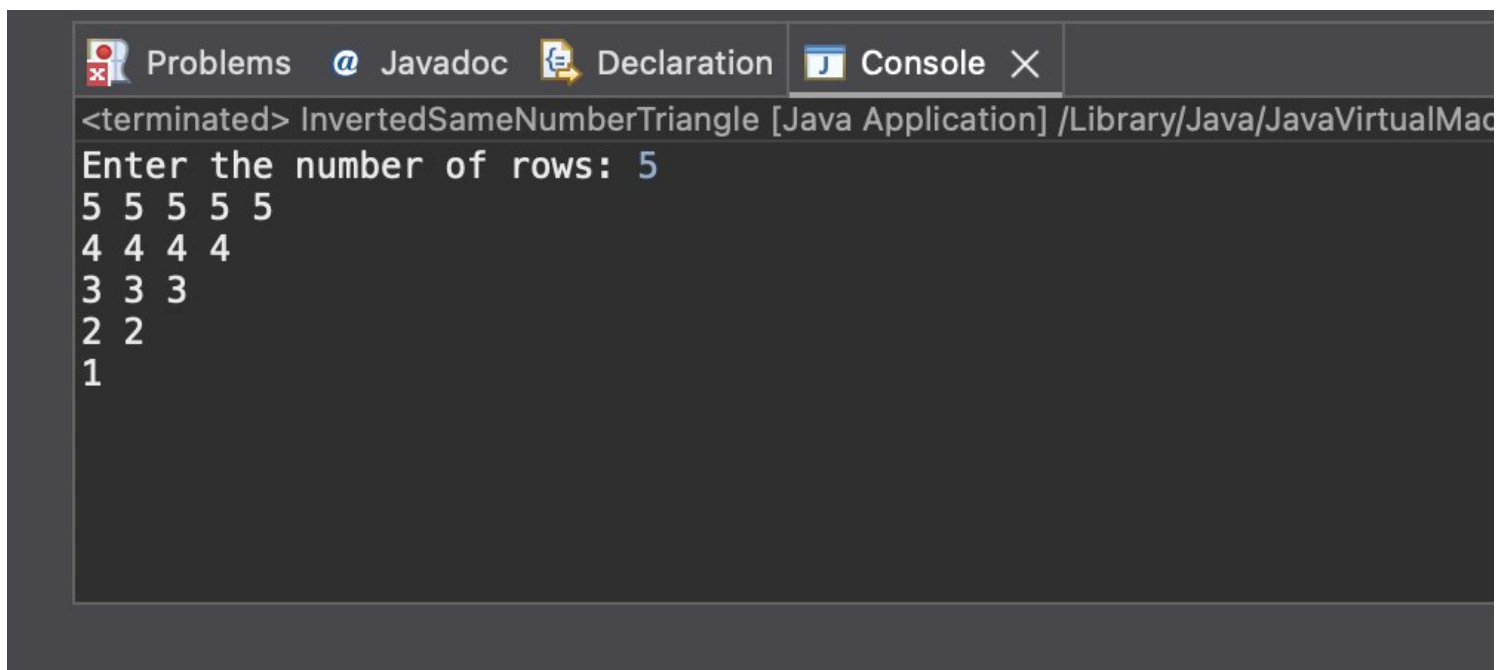
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class InvertedSameNumberTriangle {
4
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the number of rows: ");
9         int n = sc.nextInt();
10
11         for(int i = 1; i <= n; i++) {
12
13             for(int j = 1; j <= (n - i + 1) ; j++) {
14                 System.out.print( (n - i + 1) + " ");
15             }
16
17             System.out.println();
18         }
19
20         sc.close();
21     }
22 }
23
24
```

ii. Program's Output :-



```
<terminated> InvertedSameNumberTriangle [Java Application] /Library/Java/JavaVirtualMac
Enter the number of rows: 5
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```